

LDR

DETECT

LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT
How dark it is



DECIDE
Is it darker
than dusk?



DO
Switch the
lamp on



Biscuit tin alarm

DETECT
Light inside
the tin



DECIDE
Is the lid
open?



DO
Sound the
alarm



Window
Light lands
here

Track
Zigzag sensor

Legs
Either way
round

Code

● Raspberry Pi

```
from gpiozero import LightSensor  
  
ldr = LightSensor(18)  
  
light = ldr.value
```

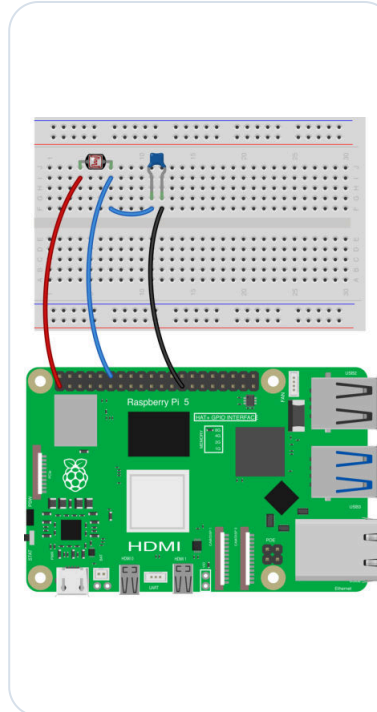
● Raspberry Pi Pico

```
from picozero import AnalogInputDevice  
  
ldr = AnalogInputDevice(26)  
  
light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. A Pi also gives you `ldr.light_detected` and `ldr.when_light`; a Pico gives you `ldr.voltage`.

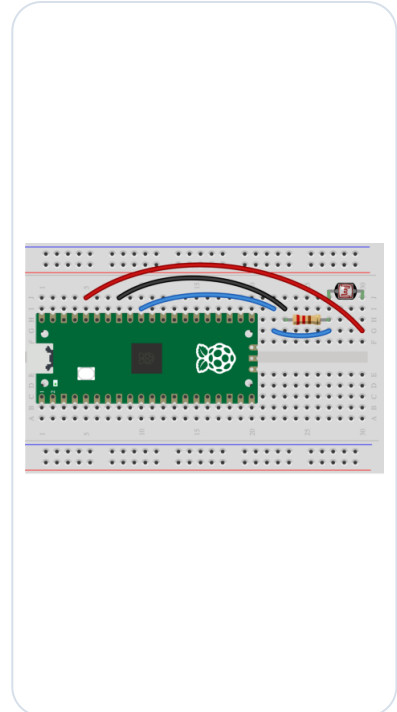
Wiring

RASPBERRY PI 5



3V3 → LDR → GPIO 18 → 1 μ F → GND

PICO



3V3 → LDR → GP26 → 10 k Ω → GND